A. Data, representations and evaluation

This supplement supports RQ3. The shared numbering is RQ1 targeted/general augmentation, RQ2 detection recovery and RQ3 feature comparison. Executable RQ1/RQ2 contributions are still required.

A.1. Images and fixed folds

Of 78,000 A–Z Kaggle ASL Alphabet training images, 60,251 original images have one accepted hand (77.24%). Failed detections are excluded. Supplied test images duplicate training sources and are not independent. Detector thresholds for the original feature cache were not retained; equal image counts do not establish equivalence with the separately recorded RQ2 settings.

Five fixed folds separate six blocks of consecutive file numbers, not verified participants or sessions. Every representation shares source IDs, labels and test folds. Each source receives one out-of-fold prediction. Folds were inspected during development; this is exploratory validation, not an untouched final test.

A.2. Four blocks, eight representations

Table A.1. D is always retained; S, E and J are independently included or omitted. Appendix B compares the eight combinations.

Block	Inputs	Meaning
D	190	Normalized image-landmark distances following Shin.
S	630	Direction cosines for 210 landmark pairs relative to image axes; not internal joint angles.
E	70	One-hot values for 19 own checks: flexion, spreading, crossing, thumb shape and contact (Appendix C).
J	15	Fourteen internal angles and one signed thumb deviation from estimated world points. MCP measurements are geometric approximations.

DSEJ has 905 inputs. E uses measured categories and a fixed column order, not the true letter. All eight representations use originals without augmentation or right normalization. D/S/E/J were recomputed on 390 sources; all 19 one-hot groups were checked across the full cache.

A.3. Fitting and reporting

Eight representations, two learners and five folds yield 80 fold models. SVM uses RBF, $C = 10$, `gamma=scale` and balanced classes. LightGBM uses 100 trees, 31 leaves, learning rate 0.1, minimum 20 examples per leaf and seed 42. Only J is median-imputed and standardized using the training fold; D/S/E pass through unchanged. No representation-specific tuning is performed. Fixed model geometry limits the interpretation.

Pooled macro-precision and macro-F1 average A–Z equally; undefined letter scores are zero. Paired analysis counts corrected and newly introduced errors on the same sources. The separate image-coordinate J controls share sources, folds and profiles. D/S use image points; E/J use world points. Appendices D–F retain their own 835-input DSJ reference, distinct from DS with 820 inputs in the main comparison.

Code and evidence: `research/feature_study.py` and `results/feature_factorial_01/`. Protocols, source hashes, predictions and replay verify computation, not anatomy or linguistic ground truth.

A.4. Outstanding RQ1/RQ2 material

RQ1 requires the targeted/general augmentation recipe, unique source counts, fixed test IDs, repeats and results. RQ2 requires executable recovery code and a visual assessment of recovered points, alongside its supplied dilation/erosion/Canny results. Keep detector/model settings controlled and fit preprocessing within training data. Derivatives are not new independent sources.

B. Eight representations and feature dependence

D denotes distances, S Shin directions, E own categories and J world-joint measurements (Appendix A). Every combination uses the same 60,251 accepted sources and five folds.

Table B.1. Pooled classification. P is macro-precision (%). E contains 70 category values for 19 checks; J uses estimated world points.

		SVM		LightGBM	
Variant	Inputs	P (%)	Macro-F1	P (%)	Macro-F1
DS	820	97.388	0.973862	96.802	0.967843
D	190	94.425	0.943763	93.633	0.935508
DSE	890	97.357	0.973477	96.781	0.967633
DE	260	94.635	0.945974	93.719	0.936395
DSJ	835	97.539	0.975244	96.713	0.966920
DJ	205	95.054	0.950107	93.832	0.937659
DSEJ	905	97.389	0.973763	96.776	0.967525
DEJ	275	94.975	0.949262	93.869	0.937914

Table B.2. Paired changes relative to DS on identical images. Net is corrected minus lost. All four contexts of each addition are retained in `factor_effects.csv`.

	SVM			LightGBM		
Variant	Corrected	Lost	Net	Corrected	Lost	Net
DS	0	0	+0	0	0	+0
D	295	2,180	-1885	504	2,524	-2020
DSE	88	110	-22	116	117	-1
DE	320	2,080	-1760	495	2,465	-1970
DSJ	143	76	+67	98	146	-48
DJ	325	1,838	-1513	511	2,405	-1894
DSEJ	145	148	-3	133	150	-17
DEJ	341	1,908	-1567	506	2,389	-1883

B.1. Coordinate-source control

D/S use image points; E/J use world points. The following control calculates J from exactly Shin's image points, without aspect correction, using shared sources/folds/profiles. World J gains 67 SVM predictions versus four for image J; the angle formula alone cannot explain the world-point gain.

Table B.3. Image-coordinate J controls without E. These are additional controls, not extra factorial arms.

	SVM		LightGBM	
Variant	P (%)	Macro-F1	P (%)	Macro-F1
D + image joints	94.423	0.943797	93.790	0.937267
DS + image joints	97.411	0.974028	96.770	0.967403

E/J are compared separately and together, with and without S. The ranking concerns these fixed profiles and sources, not universally optimal descriptions.

B.2. Permutation after fitting

Seed 20261107 selects 100 held-out images per letter/fold: 2,600 per fold and 13,000 unique sources. All models share this subset, distinct from the full classification denominator. Five seeded row permutations move a feature group's columns together. Labels and fitted models stay fixed.

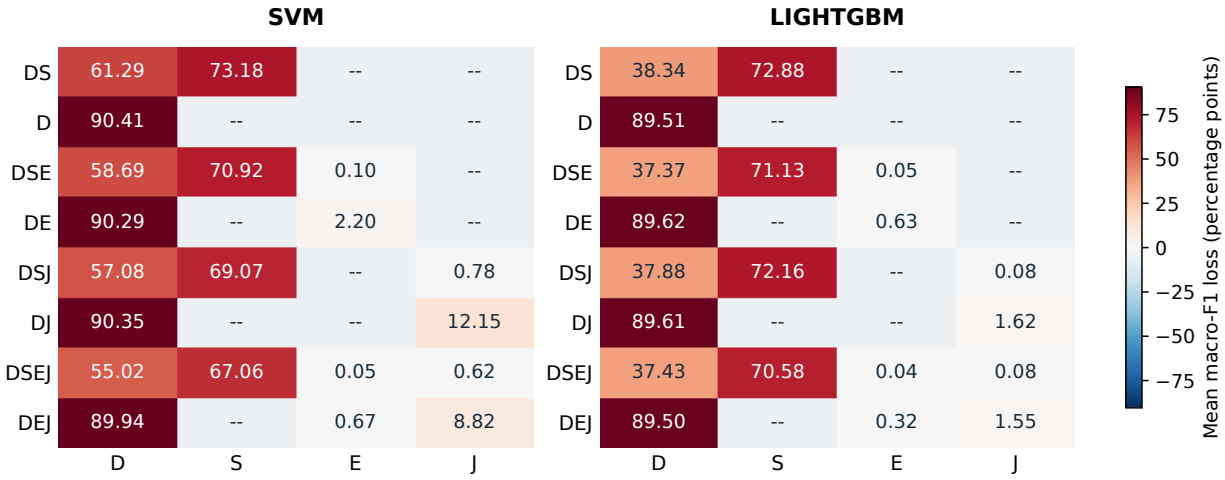


Figure B.1. Present D/S/E/J blocks in each representation. Values are mean macro-F1 losses in percentage points over five folds and five shuffles; one point equals 0.01 F1. – denotes an absent block. A positive loss indicates dependence under this perturbation.

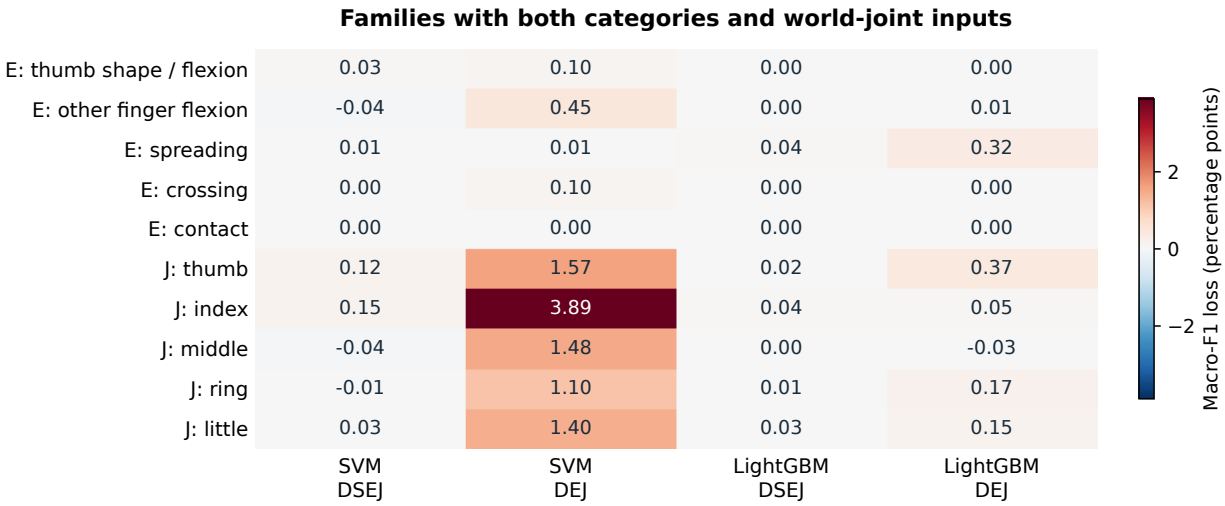


Figure B.2. Five E families and five J families when both blocks are present. Families are shuffled jointly. Related information may substitute across blocks. Negative losses are retained: those perturbations improved the mean score.

Joint column permutation preserves one-hot validity. Fold/repeat orders are shared across variants and learners. The optimized SVM route reuses equivalent RBF distances and checks original/perturbed predictions against ordinary SVC. Seeds and execution are in `protocol.json` and `execution_backend.json`. CSVs retain precision losses and descriptive spread, not an independent confidence interval.

B.3. The 19 individual checks

Individual E checks are tested in DSE and DE, without J, so direct world-joint inputs do not simultaneously duplicate their angle information. The preceding family comparison covers full representations with J.

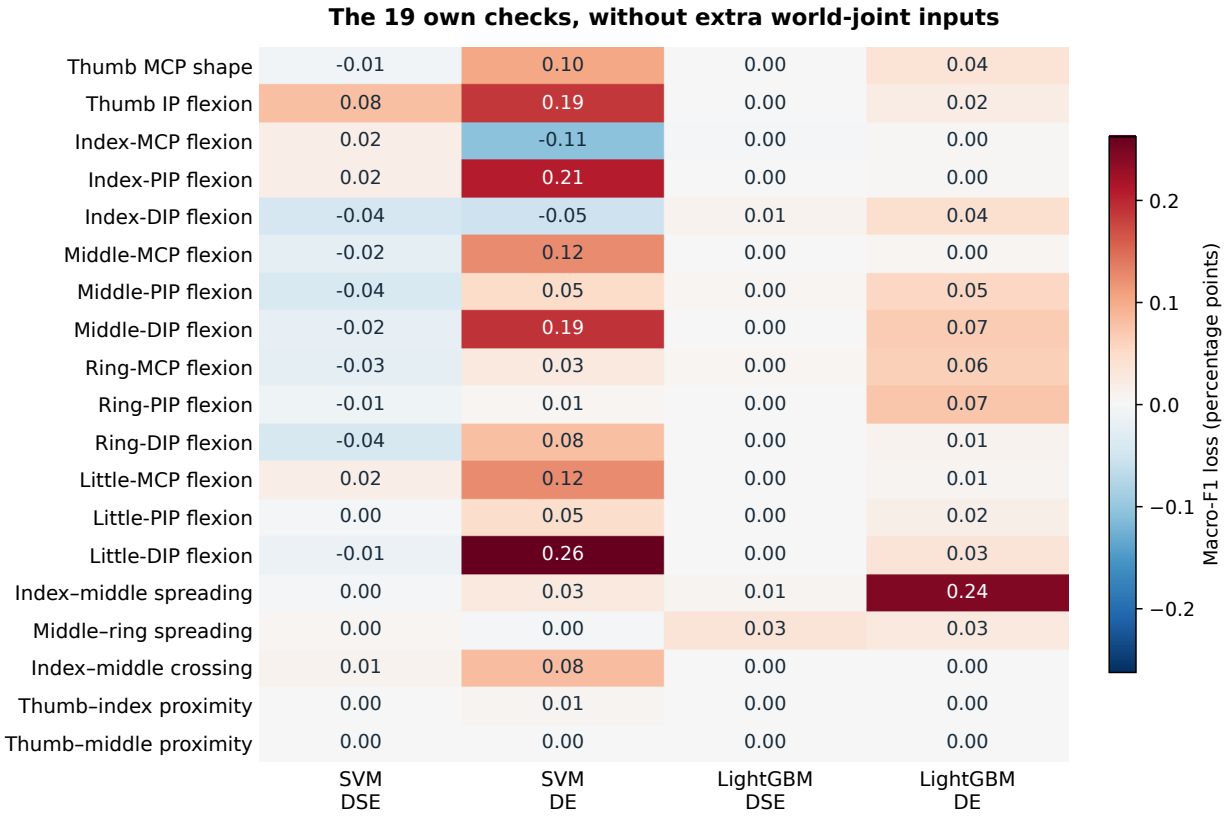


Figure B.3. Macro-F1 loss per own check, jointly permuting all category columns. DSE includes Shin directions; DE omits them. Means use the same 13,000 held-out images and five shuffles per fold. Appendix C explains the checks.

Permutation measures dependence of a fitted model under perturbation, not the gain from adding a feature to a newly fitted model. The eight training variants answer that complementary question. Related features may substitute, yielding a small individual loss; shuffling may also create unusual combinations. This is not a causal ranking of intrinsically important hand features.

Full fold/letter metrics, confusions, paired comparisons, permutation outputs and replay are in `results/feature_factorial_01/`. Computational verification does not establish anatomy or generalization to new users.

C. Hand features and tool inspection

Own categories make hand shape explicit: bent joints, separated/crossed fingers, thumb position and thumb-tip proximity. The workbook defines 19 checks: 13 flexion, two spreading, one crossing, one thumb shape and two proximity checks. They describe estimated landmarks; one check may produce multiple model inputs (Appendix A).

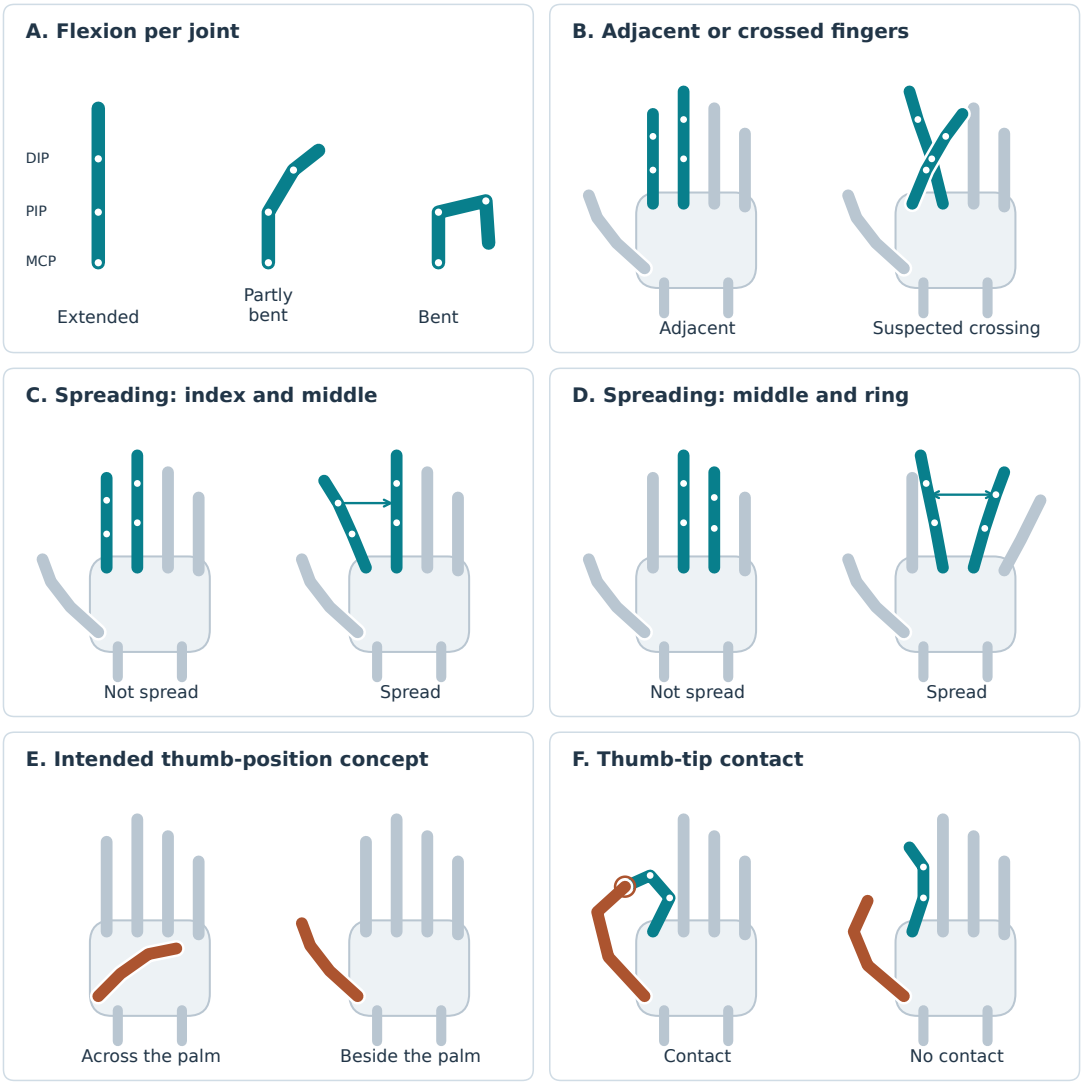


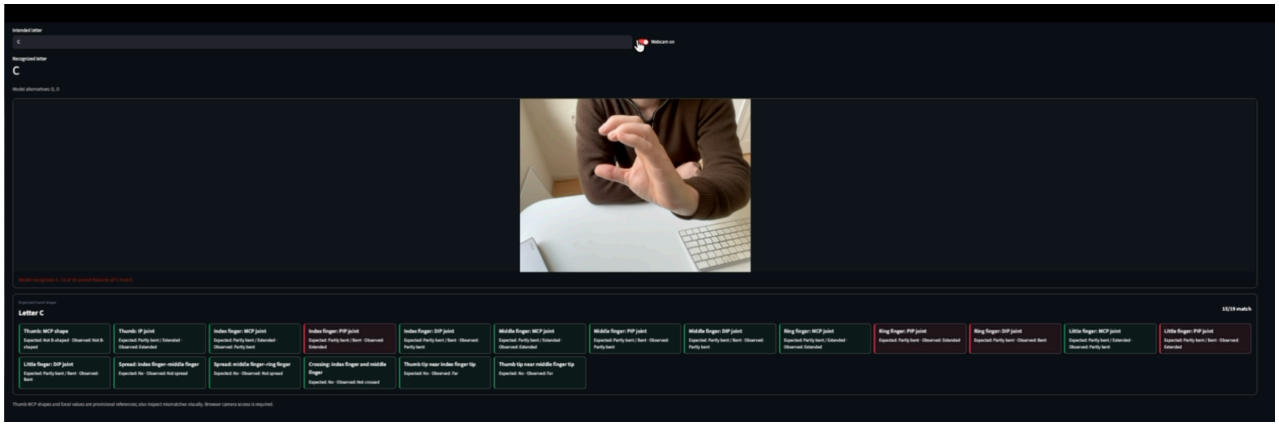
Figure C.1. Conceptual own features. Highlighted fingers identify the check. A: flexion per joint. B: adjacent/crossed index and middle fingers. C–D: spreading for two separate finger pairs. E: the intended across/beside-palm thumb distinction. F: thumb-tip proximity to index/middle separately. Drawings explain concepts; they are not measured hands or classification results.

Not spread does not imply contact. A pair failing the spreading rule may be adjacent, touching, crossed or strongly bent. Crossing has a separate rule; direct proximity is checked only for thumb-tip–index-tip and thumb-tip–middle-tip.

Thumb shape approximates the intended concept. Across-palm position is approximated by signed thumb-MCP deviation: `B-shaped` or `Not B-shaped`. It does not directly measure covered palm area; the latter category includes several thumb positions.

C.1. Using the feature tool

The target letter selects expected workbook categories but does not affect recognition. The tool compares measured and expected categories. The supplied screenshot shows intended C and predicted C, with 15/19 checks matching. Correct recognition can coexist with individual category mismatches.



Index finger: PIP joint

Expected: Partly bent / Bent

Observed: Extended

Mismatch

Index PIP: a flexion-category mismatch.

Spreading: index–middle

Expected: Not spread

Observed: Not spread

Match

Index–middle: not spread, as expected.

Figure C.2. Original screenshot with two re-typeset detail cards showing the same expected/observed values as sharp vector text. Green: match; red: mismatch; grey: not scored. These are not letter-confidence scores.

C.2. Measurement rules

An internal angle is measured between the two segments around a joint; a straight joint approaches $180°$. MCP is the knuckle, PIP the middle joint and DIP the end joint; the thumb has an IP. The 13 checks comprise thumb IP and MCP/PIP/DIP for the other four fingers. Thumb MCP uses the separate shape rule in panel E. Measurements use estimated world points.

Table C.1. Main rules from `config/thresholds.json`. Thresholds are project choices.

Feature	Rule
Finger flexion	Bent at most $130°$, extended from $150°$, partly bent between. Thumb IP uses $80°$ and $155°$.
Spreading	Index–middle and middle–ring separately: normalized gap at least 0.25, depth gap at most 0.6, positive direction cosine and PIP/DIP angles at least $120°$. Divergent directions are allowed.
Suspected crossing	Index/middle reverse lateral order along their profiles, depth gap at most 0.7 and sufficient extension. Crossing 2D lines alone is insufficient.
Thumb shape	B-shaped when signed palm-directed deviation exceeds $16°$; otherwise not B-shaped.
Thumb proximity	Tip distance divided by wrist–middle-MCP length: contact at most 0.1, near up to 0.825, far above. Both thumb-tip pairs are measured separately.

The workbook may allow several categories. For no contact, both near and far count as matching. Unknown measurements are not confirmed agreement; blank references are not scored. Green confirms the chosen rule, not anatomy or ASL execution.

D. Exploratory left/right checks: mirror training

All 78,000 original images and their horizontal RGB reflections were independently detected in IMAGE mode: at most two hands, detection/presence thresholds 0.5. There are 60,251 accepted originals and 60,201 accepted mirrors. Each presentation uses its own returned landmarks and hand label; anatomical indices are unchanged.

The SVM reference fits accepted original sources; the candidate adds accepted mirrors of the same fitting sources. Each source retains total weight one. Imputation/scaling, numeric gamma and original class weights are shared. Source and derivative stay in the same fold. Primary classification scores exclude failed detections, as in the feature comparison.

Seed 20261106 selects half the sources per letter/fold for mirroring: 39,000 original and 39,000 mirrored presentations, before checking detection success. The fixed mixture has 60,225 accepted detections. Models share accepted sources within each presentation. The mixture is a different test distribution from originals alone.

Table D.1. SVM classification conditional on accepted detection. N is the shared accepted-image denominator for each presentation; P is macro-precision. All-attempt scores remain in the result files.

Test view	Training	N	P (%)	Macro-F1	Correct
Original	Original	60,251	97.539	0.97524	58,894
Original	+ mirroring	60,251	97.495	0.97493	58,868
Mirrored	Original	60,201	26.007	0.19249	14,027
Mirrored	+ mirroring	60,201	97.293	0.97250	58,676
Fixed mixture	Original	60,225	73.566	0.62650	36,452
Fixed mixture	+ mirroring	60,225	97.451	0.97421	58,801

On the mixture, mirror training gains 22,349 correct predictions; on originals it loses 26. Mirror pairs share person, pose and background and are not independent left-handed participants. Detector-Left/Right is not independently verified ground truth.

D.1. Selected reflected examples

The original is left and its independently detected reflection right. Their source is excluded from fitting the corresponding fold.

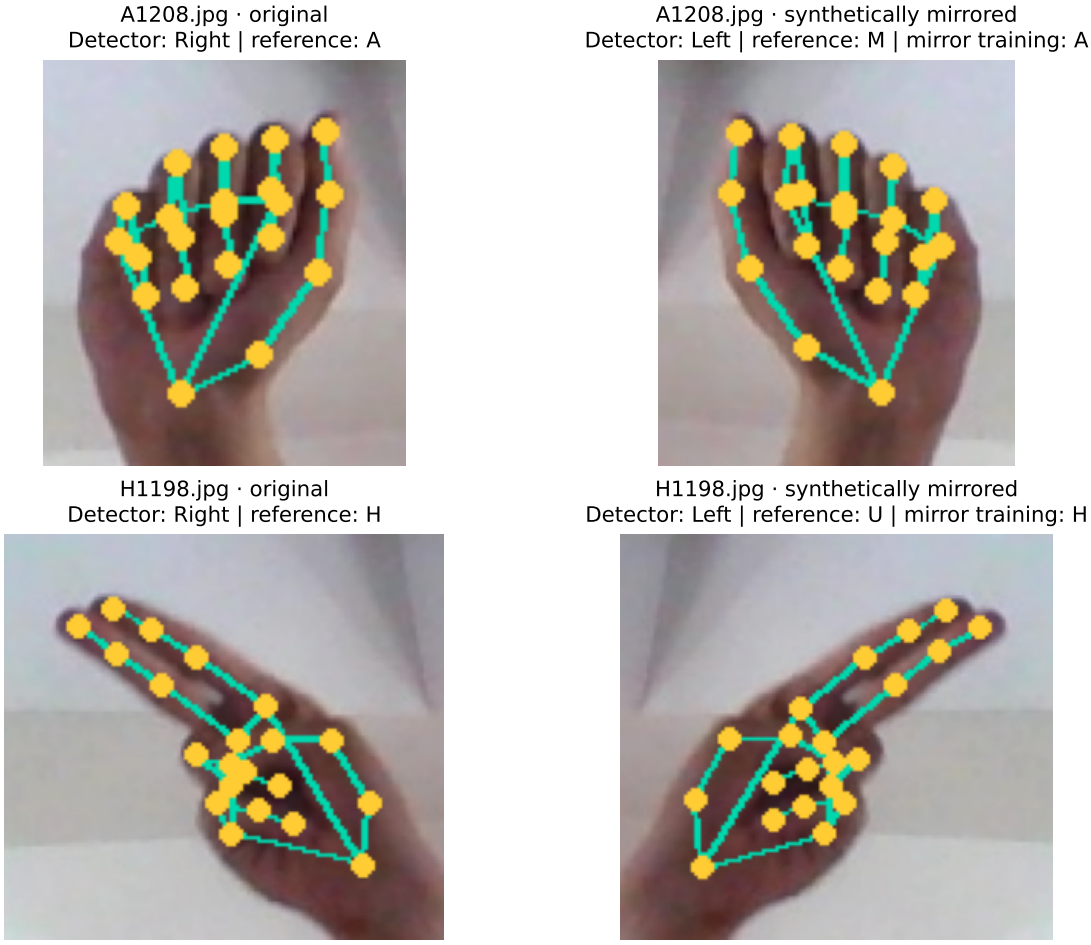


Figure D.1. Selected A1208 and H1198 examples. Without mirror training, reflections are M and U; with mirror training, A and H. Predictions are from fold 5, which excludes these sources from fitting. These are illustrations, not an independent test.

For each A/H, selection takes the first matching source ID with valid Right/Left detections, a correct original, an incorrect reflection without mirror training and recovery with it. This estimates no success rate. Reflection does not turn a visible palm into a visible hand back. Predictions and settings: `results/mirror_01/`; selection/hashes: `results/appendix_visuals_01/`.

E. Exploratory G/H rotation checks

Motivation and hypothesis. For usual ASL G/H, the palm faces inward or sideways, as described by Bill Vicars, ASL University. A camera facing the signer therefore sees mainly the back or side. The selected Kaggle examples in Figure E.1 show the palm. Shin used this dataset, so a derived recognizer may learn a different view. Camera viewpoint, hand identity and all G/H labels have not been independently checked. This motivates a possible training/use mismatch, not a demonstrated fault in Shin's learning method.

We add only $180°$ camera-y G/H fitting rotations while retaining originals, without mirror training, webcam calibration or routing rules. Both arms are ordinary 26-class SVMs with shared sources, folds, original preprocessing, numeric gamma and class weights.

The 210 pair directions change as $(d_x, d_y, d_z) \mapsto (-d_x, d_y, -d_z)$. This rotation has determinant $+1$, unlike reflection. Distances and intrinsic joint measurements remain equal. It is equivalent to wrist-centred image-landmark rotation, but creates no new RGB view, hidden surfaces or detector errors.

The fixed population has 60,251 accepted sources: 2,430 G and 2,371 H. G/H originals and one rotated copy weigh 0.5 each; other sources weigh one. The full fit has 65,052 rows. Ordinary held-out images remain unchanged. A freshly fitted reference exactly reproduces the stored out-of-fold predictions.

Table E.1. Pooled results on 60,251 unchanged accepted test sources. P is macro-precision over A–Z. Scores condition on accepted original detection.

Training	Acc. (%)	P (%)	Macro-F1	Correct
Original	97.74776	97.53902	0.97524391	58,894
+ G/H rotation	97.74610	97.53874	0.97523305	58,893

Four errors are corrected and five correct predictions lost: net -1. Fold nets are $+2$, 0, -1, -1, -1. There is no general improvement on the original view. These sources were inspected during development; no confirmatory significance claim is made.

Table E.2. G/H within the 26-letter task; false positives from other letters count. This is not a binary G/H classifier.

Letter	Training	N	Precision	Recall	F1
G	Original	2,430	0.989353	0.994239	0.991790
G	+ rotation	2,430	0.987735	0.994239	0.990976
H	Original	2,371	0.986441	0.981864	0.984147
H	+ rotation	2,371	0.986023	0.981864	0.983939

G/H recall stays equal while precision/F1 decline slightly. The original test checks regression on the dataset view, including palm-facing G/H. It does not independently validate ASL execution. Recognizing an atypical palm-facing execution need not be preserved if the goal is usual ASL orientation. An independently reviewed real hand-back test is still needed. Rotation angle/axis were chosen after exploratory work.

E.1. Rotated landmarks from real fitting sources

Figure E.1 shows G1128/H1198 and the 21 stored points used for feature extraction. Left is the source photo; centre is the original skeleton; right is its half-turn. No redetection is performed.

Wrist 0 is the pivot. For image point (x_i, y_i, z_i):

$$(x_i', y_i', z_i') = (2x_0 - x_i,\ y_i,\ 2z_0 - z_i).$$

The dashed line is the camera-y axis. Wrist-relative x/depth z reverse sign; y is unchanged and points down. All skeletons share a scale and are centred on the hand. These are normalized image coordinates with estimated depth, not metres.

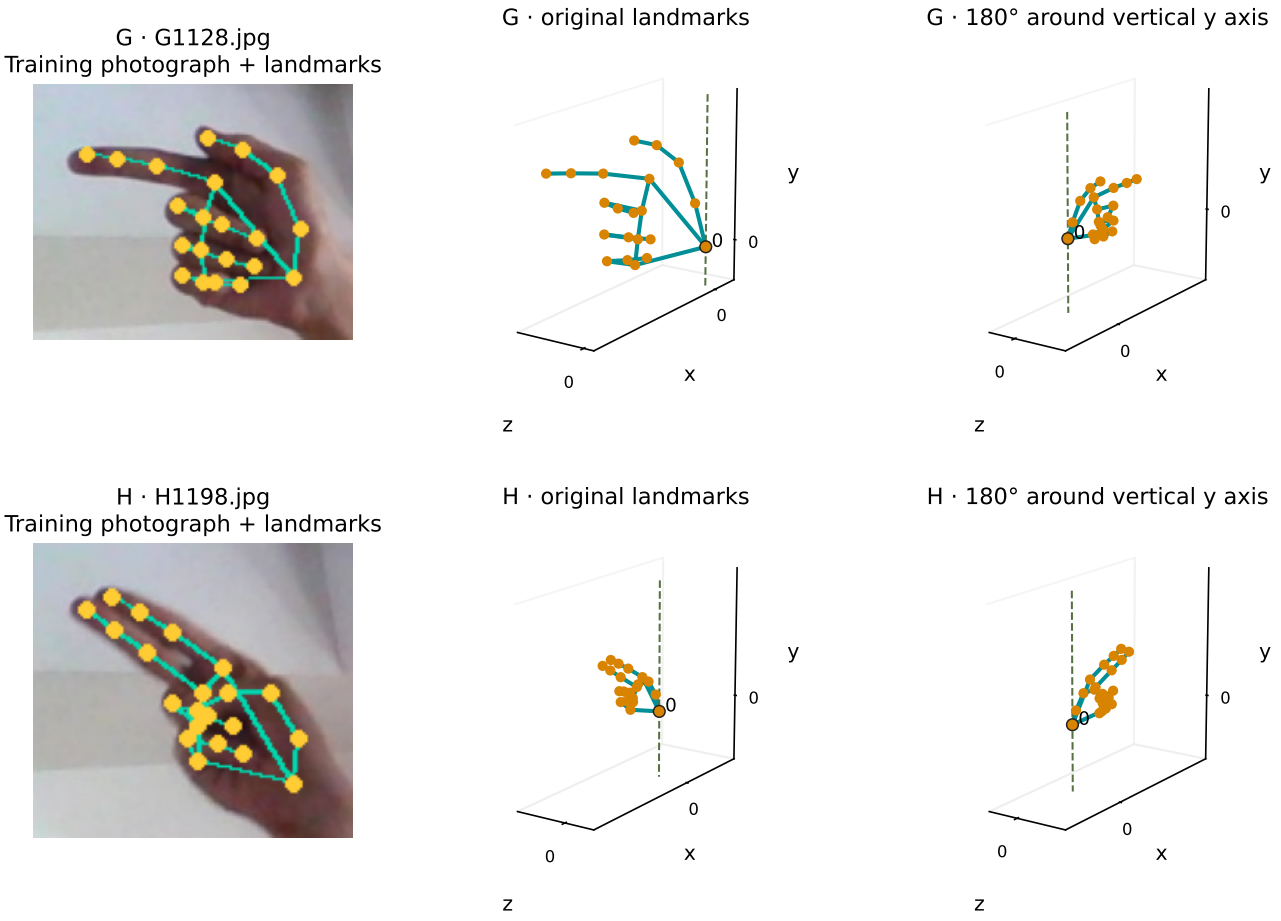


Figure E.1. Two actual fitting sources: source photograph, original points and equivalent 180° vertical-axis rotation through the wrist. Orange marks landmarks; black marks the wrist. Right panels contain transformed geometry, not new rendered camera images.

Recomputed Shin features match the actual augmentation for both examples within 10^{-6}. The 190 distances and 15 joint measurements are unchanged. Source IDs, coordinates and checks are in `results/gh_training_rotation_01/`.

E.2. Occlusion and uncertain landmarks

Two distinct approximations. Rigid rotation retains all 21 points. To approximate occlusion, we build a coarse volume around source points: a palm ellipsoid and tapered rounded finger segments. A depth buffer determines the frontmost component. Width, thickness, colour, light and background are fixed assumptions, without anatomical fitting or a new photograph. Geometry uses no letter label. The assumption that every source shows a palm was not independently checked.

MediaPipe accepts all 64 predefined source photographs (32 G/32 H), but no render at either zero or 180 degrees: 0/128 synthetic images. Detection/presence thresholds are 0.5. This route supplies no new detector landmarks for fitting; failed detections are not replaced by known source points.

Separate robustness proxy. All 4,801 G/H sources receive wrist-centred image/world rotations. Gaussian noise is added to PIP/DIP/IP/thumb points behind a different estimated hand component. Fixed standard deviations are 2% or 5% of wrist–middle-MCP length per coordinate. Both coordinate sources share the random pattern at their own scale. Other points, wrist and knuckles remain exactly rotated. This assumes point uncertainty, not a measured distribution of MediaPipe occlusion errors; within-finger self-occlusion is not modelled.

The strengths share a reproducible pattern per source. All 835 features are recomputed. Zero noise matches the existing rotation on every G/H source within 10^{-5}. Original/copy weights remain 0.5. Folds, original preprocessing, numeric SVM gamma and class weights stay fixed. Only fitting sources produce fitting copies; no webcam captures enter training.

Table E.3. Regression check on 60,251 original accepted tests. P is macro-precision (%). Corrected/lost counts are paired against original training. This is not a real hand-back test.

Training	P (%)	Macro-F1	Correct	Corrected/lost
Original	97.53902	0.975244	58,894	0 / 0
Rotation only	97.53874	0.975233	58,893	4 / 5
Rotation + 2% point noise	97.53874	0.975233	58,893	4 / 5
Rotation + 5% point noise	97.53764	0.975219	58,892	3 / 5

The 2% arm exactly matches pure-rotation predictions on originals; the 5% arm loses one additional correct prediction. Neither adds an original-view gain. This does not validate the noisy points as realistic hand-back observations.

Additional artificial test. Only held-out G/H landmarks are rotated; the other 24 letters remain unchanged. Models are not refitted. This tests whether the offered geometry is learned, not real hand-back detection.

Table E.4. Artificial test: rotated G/H landmarks, other letters unchanged. The same 60,251 sources; no new RGB detections.

Training	P (%)	Macro-F1	Correct
Original	85.0689	0.869284	54,151
Rotation only	97.5861	0.975715	58,924
Rotation + 2% point noise	97.5830	0.975683	58,922
Rotation + 5% point noise	97.5802	0.975653	58,920

Pure rotation gains 4,773 correct predictions over original training. Rotated G/H recall rises from 0% / 0.04% to 99.47% / 99.45%. Point noise adds no gain. Rotation learns the offered geometry; the almost unchanged original score checks preservation of the original view.

E.3. Illustrating the occlusion assumptions

The source photographs match Figure E.1. Volumes use their stored points. Red circles mark positions behind a different component. Their location remains known from the source; a real detector would estimate it again.

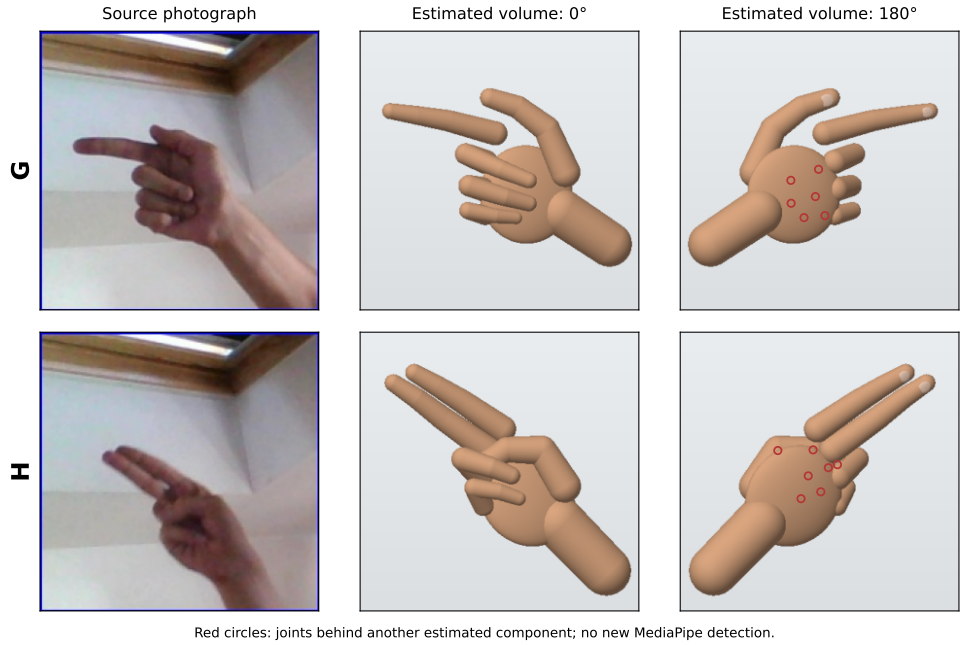


Figure E.2. Two fitting sources and approximate volumes. Red circles mark points behind a different hand component. These illustrate assumptions, not new observations; MediaPipe accepts none of these renders.

Protocols, source selection, folds, letter metrics, predictions and hashes are in `results/gh_occlusion_01/`. Both strengths are exploratory; neither is selected as a new deployment default. The intended ASL execution still requires an independently reviewed hand-back evaluation including failed detections.

E.4. Selected free live-tool examples

The user saved four examples during free testing with visible predictions and model selection. All have accepted detection. Stored points are replayed through the 835-input reference and G/H rotation model. Saved model hashes, predictions and input vectors are checked; no redetection or fitting occurs.

Table E.5. Identical stored landmarks through both models. Intended letters are user annotations. View labels were assigned visually, not by an independent ASL expert.

Example	Intended	Visible view	Reference	G/H rotation
A	G	Back visible	M	H
B	G	Back visible	Z	H
C	H	Back visible	Z	H
D	G	Palm visible	G	G

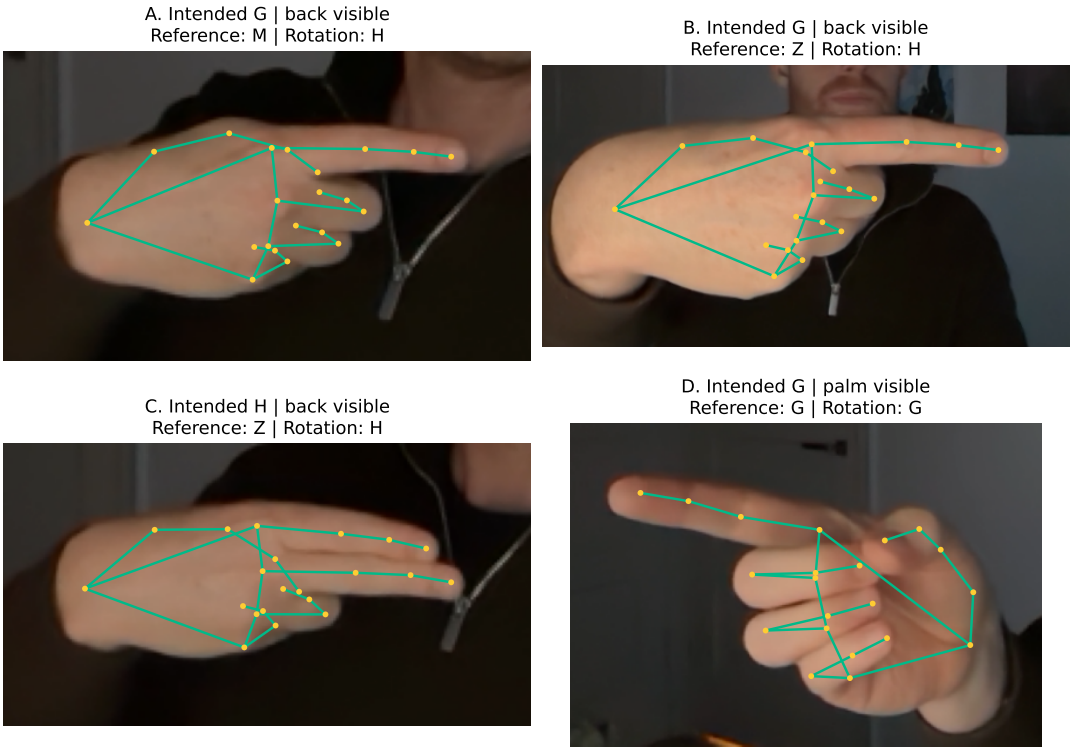


Figure E.3. All four new captures, cropped with their original landmarks. A/B: intended G remains H; C: H recovered; D: G recognized with palm visible. Reference/Rotation are model outputs, not confidence estimates.

Difference from the artificial test. Rigid rotation preserves intrinsic joint geometry. A/B instead have estimated middle DIP angles 161.7 and 164.7 degrees, compared with median 113.2 in 2,430 G fitting sources (middle 50%: 109.2–118.2). Signed thumb deviations are -1.9 and 2.7 degrees versus median 35.9. D measures 120.0 and 29.7 degrees respectively. These are world-point estimates, not true 3D anatomy. Pose, viewpoint and detector error cannot be separated here, nor does the comparison establish one angle as the cause of misclassification.

The high artificial score does not directly transfer: H is recovered but both hand-back Gs remain H. Rotation is not a demonstrated solution to the observed live confusion. Four selected images do not estimate representative precision, F1 or error frequency. Sources, measurements and hashes: `results/gh_live_diagnostics_20261107/`.

F. Exploratory left/right checks: normalization

We compare no change, mirror training and normalization to detector-Right before fitting/inference. Detector-Left image points receive $x' = 1 - x$ and world points $x' = -x$; y, z, anatomical indices and the letter label stay unchanged. The hand label becomes Right. Detector-Right is unchanged. This is a post-detection reflection, not RGB redetection or palm/back rotation.

F.1. Fixed design and controls

Sources, independently detected presentations, folds and mixture are exactly those of Appendix D. Only accepted original sources enter fitting. Of 60,251 accepted originals, the detector labels 2,827 Left and 57,424 Right. Normalization adds no fitting rows; each source keeps weight one. Both RGB presentations are accepted for 58,941 sources; 4,261 retain the same hand label across views.

The 835-input representation follows DSJ. For Left, only the 210 x-direction cosines change sign; distances and hand-aware joint measurements remain equal. On 312 detections, this cache transformation exactly matches reflecting points then recomputing features, as in the live model. Unknown hand labels are rejected.

All variants share original-fitting preprocessing. SVM keeps $C = 10$, numeric gamma and reference class weights; LightGBM keeps the fixed Appendix A profile. There is no extra tuning. Sources and derivatives remain excluded together from the corresponding test fold. The idea followed earlier inspected outcomes and is exploratory.

Table F.1. Accepted detections only; each presentation shares N across variants. Failed detections are excluded. The primary orientation reference is the 835-input SVM on originals. LightGBM is shown compactly on originals.

Model	Test view	Training/input	N	P (%)	Macro-F1	Correct
SVM	Original	No change	60,251	97.539	0.97524	58,894
SVM	Original	Mirror training	60,251	97.495	0.97493	58,868
SVM	Original	Right normalization	60,251	97.322	0.97320	58,758
SVM	Mirrored	No change	60,201	26.007	0.19249	14,027
SVM	Mirrored	Mirror training	60,201	97.293	0.97250	58,676
SVM	Mirrored	Right normalization	60,201	94.885	0.94802	57,239
SVM	Fixed mixture	No change	60,225	73.566	0.62650	36,452
SVM	Fixed mixture	Mirror training	60,225	97.451	0.97421	58,801
SVM	Fixed mixture	Right normalization	60,225	96.076	0.96047	57,987
LightGBM	Original	No change	60,251	96.713	0.96692	58,454
LightGBM	Original	Mirror training	60,251	96.990	0.96962	58,581
LightGBM	Original	Right normalization	60,251	96.490	0.96459	58,304

On originals, SVM corrects 157 errors and loses 293 correct predictions relative to no change: net -136. Fold nets are -51, -53, -28, -3, -1. Relative to mirror training, the net difference is -110. Normalization does not improve original-image classification.

The separate mixture has 60,225 accepted detections. Its SVM macro-F1 changes from 0.62650 to 0.96047; mirror training gives 0.97421. This is a different test distribution. Excluding failed detections leaves precision unchanged here, because those attempts predict no letter and create no false positive. F1 changes as their false negatives are excluded. All-attempt scores remain available separately.

F.2. Interpretation and live use

Normalization remains below the original reference on originals and below mirror training on the mixture. Detector-label errors and asymmetric detections are possible explanations, but their effects were not isolated. Labels are unverified, and mirror pairs are not independent participants. Evidence: `results/handedness_01/`. The full-data deployment fit is not an independent test.